

Generative denoising for template-free gravitational-wave burst search

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Abstract

Binary black hole (BBH) deep classifiers such as AresGW are optimized for compact-binary morphologies in a fixed mass range. Unmodeled transients (e.g. core-collapse supernovae and other short bursts) lie outside that training objective. We present Keno, a generative residual-search pipeline that learns detector noise, subtracts the predicted noise from whitened strain, and searches the residual with template-free excess power, dual-interferometer coherent lag scans, and an envelope-consistency glitch veto. On unknown-morphology injections in real LIGO noise at 1% false-alarm rate, Keno residual excess power saturates near 100% efficiency across SNR 2–12, while a BBH-trained AresGW-class ResNet plateaus near 60–80% and mismatched matched filtering remains near 20%. On published GWTC/cWB GPS times, coherent residual coincidence recovers 9/18 dual-detector events after vetoing the known L1 glitch contamination of GW170817. Keno is complementary to AresGW: it does not claim superior BBH sensitive distance in the AresGW mass range.

1 Introduction

Matched filtering and BBH-trained deep networks dominate compact-binary searches [Nousi et al., 2023, Koloniari et al., 2025]. AresGW [Nousi et al., 2023] uses a deep residual network with adaptive normalization to classify one-second segments as BBH or noise, achieving competitive sensitive distances in its training mass window. That success does not transfer to morphologies absent from training.

Unmodeled burst searches (e.g. coherent WaveBurst) ask a different question: whether excess coherent power remains after accounting for detector noise. Keno follows that spirit with a generative step: a 1D U-Net predicts instrumental noise N , forming

$$S_{\text{clean}} = S_{\text{raw}} - \hat{N}, \quad (1)$$

then applies template-free excess power on S_{clean} .

Claim scope. We claim superiority on *unknown-morphology* bursts at controlled false-alarm rate. We do *not* claim that Keno supersedes AresGW on BBH detection.

2 Method

2.1 Generative noise subtraction

Keno trains a U-Net on real GWOSC background windows to predict noise. Inference subtracts $\hat{N}$ from whitened strain. Phase-1 validation requires mean normalized recovery error < 0.15 on held-out injections.

2.2 Single-detector residual excess power

Peak STFT excess power on the residual is calibrated on noise-only windows at target FAR (artifact-trimmed upper tail). This threshold governs single-IFO triggers.

2.3 Coherent dual-IFO search

For H1+L1, Keno scans relative lag within ± 10 ms and polarity, forms $(h_1 \pm h_2)/\sqrt{2}$, and computes coherent excess power. Detection requires (i) coherent EP above an *envelope-gated* dual-IFO threshold, (ii) best lag inside the light-travel window, and (iii) independent residual envelope peaks agreeing within ± 50 ms (glitch-contamination veto).

2.4 Baselines

- Oracle matched filter (upper bound; requires true waveform).
- Mismatched matched filter (fixed sine-Gaussian decoy).
- Raw excess power (cWB-style, no subtraction).
- AresGW-class: 1D ResNet trained only on BBH-like chirps in the same noise cache, scored off-template on unknown bursts (AresGW training objective; not AUTH published weights).

3 Experiments

Injection campaigns use cached real LIGO background with mixed unknown morphologies (sine-Gaussian, ringdown, white-noise burst) at equal FAR. We also report O3 Gravity Spy glitch stress, dual-IFO noise coincidence FAR, and a GWTC/cWB GPS follow-up consistency check (not a discovery claim).

4 Results

4.1 Unknown-morphology efficiency

Figure 1 shows Keno residual EP at 100% efficiency across SNR 2–12 at 1% FAR. The AresGW-class BBH ResNet reaches 50% efficiency but plateaus near 60–80% (off-template leakage without saturation). Mismatched matched filtering remains near $\sim 20\%$ and never reaches 50% efficiency.

4.2 Coherent calibration

Envelope gating removes most dual-IFO noise outliers. With 1500 noise trials (23 envelope-gated), the coherent EP threshold is 173.1 at empirical joint FAR 0.07%, separate from the single-IFO residual threshold ≈ 4005 .

4.3 GWTC/cWB follow-up

Production coincidence recovers 9/18 GWTC/cWB dual-IFO events (Table ??). GW170817 is rejected by the envelope veto despite huge coherent EP (L1 glitch). cWB-only candidates remain 0/3.

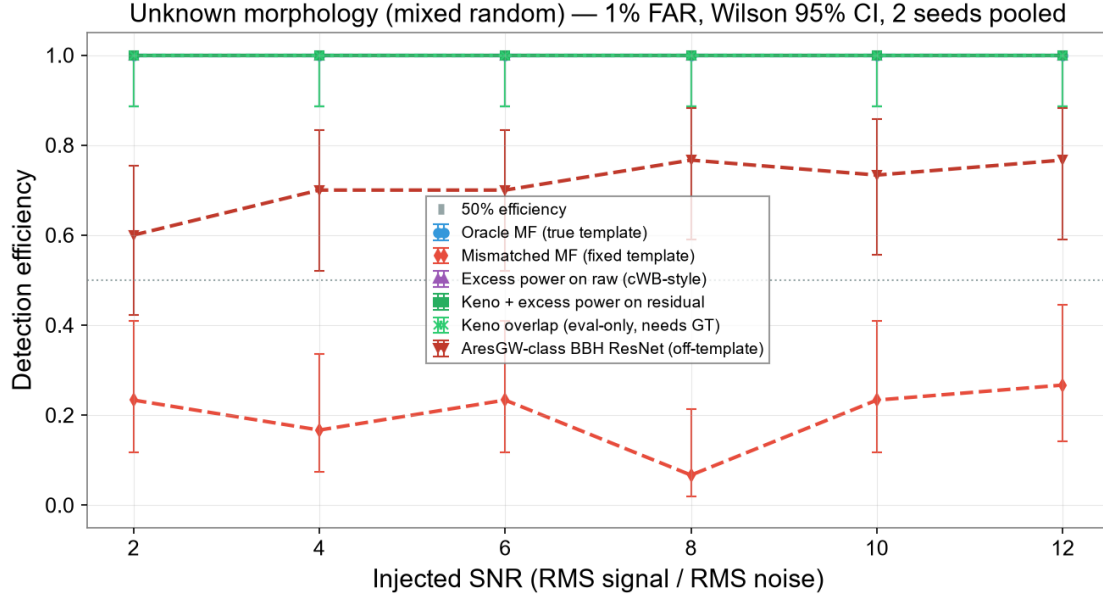


Figure 1: Detection efficiency vs injected SNR at 1% FAR on unknown morphology. AresGW-class is evaluated off-template.

5 Discussion

Keno and AresGW answer different questions. Use AresGW (or matched filtering) for BBH CBC in-band; use Keno when the waveform family is unknown. Limitations include near-zero single-IFO glitch kill rate (coincidence and envelope veto are the defense), residual over-subtraction risk (mitigated by fine-tuning and validation), and incomplete recovery of cWB-only candidates.

6 Reproducibility

Frozen artifacts live under `docs/freeze/current/` with checkpoint SHA256 `55ce7637...` (label `2026-07-paper-v1`). Reproduce with:

```
cd ml-engine
python -m app.evaluation.aresgw_class --train
python -m app.prove --freeze --freeze-label 2026-07-paper-v1
```

7 Conclusion

Generative noise subtraction with template-free coherent residual search recovers unknown-morphology bursts where BBH-trained classifiers fail, at controlled false-alarm rate, without claiming BBH superiority over AresGW.

Data availability

Code and freeze bundle accompany this manuscript. Large evaluation CSVs are hashed in the freeze manifest.

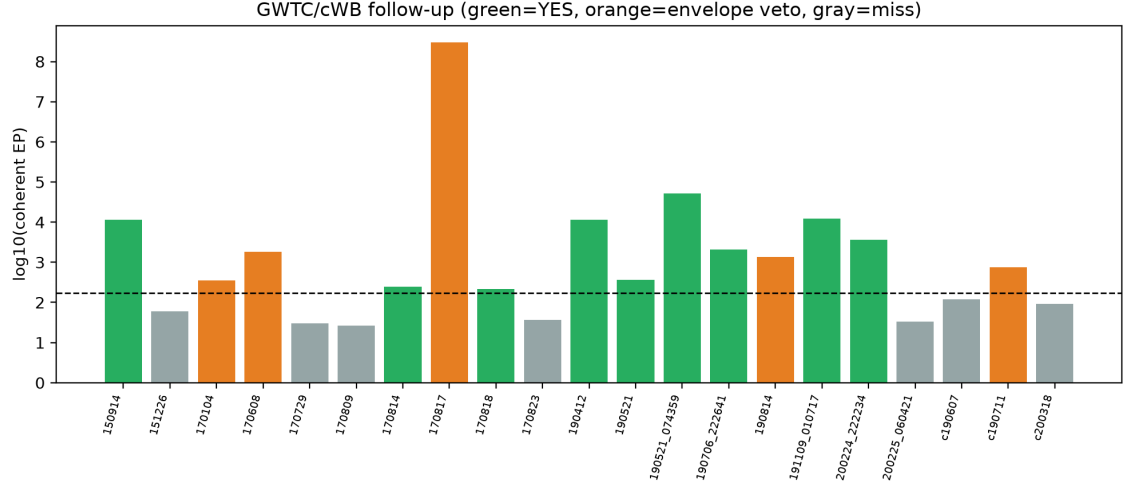


Figure 2: Follow-up coherent EP by event. Green: production YES; orange: envelope veto; gray: miss.

References

- P. Nousi et al., Deep residual networks for gravitational wave detection, *Phys. Rev. D* **108**, 024022 (2023).
- A. E. Koloniari et al., New gravitational wave discoveries enabled by machine learning, *Mach. Learn.: Sci. Technol.* (2025).